Debug Reference Document

DATA ALIGNER

Date: Fri May 08 2026

Bug ID	Error / Symptom	File · Location · Fix Applied
BUG-01	Syntax error near 'if' agent_apb/cfs_apb_driver.sv(44)	`agent_apb/cfs_apb_driver.sv  Line: 44` **Fix:** Added missing semicolon before the if statement.
BUG-02	Fatal error in cfs_algn_env:: connect_phase Line: 129	`env/cfs_algn_env.sv  Lines: 100, 129` **Fix:** Interchanged rx and tx agents at line 129. Uncommented line 100. Added default map for predictor.map.
BUG-03	Unknown values on APB interface signals at 3 ns	`apb_driver / model/cfs_algn_model.sv` **Fix:** Fixed APB driver idle, setup and access phase signal assignments.
BUG-04	UVM_ERROR scoreboard(271) @685ns RX mismatch: expected CFS_MD_ERR, received CFS_MD_OKAY	`scoreboard/cfs_algn_scoreboard.sv  Line: 271 model  Line: 185 cfs_rx_ctrl.v Lines: 84-89` **Fix:** Corrected error-transfer formula in model at line 185. Changed pop_back to pop_front in 2 scoreboard functions. Fixed if-condition for md_rx_err in cfs_rx_ctrl.v lines 84-89.
BUG-05	UNKNOWN_VALUE_VALID_A assertion at 35 ns on md_tx_if (cfs_md_if.sv line 215)	`interface/cfs_md_if.sv  Line: 215 cfs_tx_ctrl.v` **Fix:** Removed '& pop_ready' from pop_valid: assign pop_valid = !(pop_empty); Removed '-1': assign push_full = (push_fifo_lvl == FIFO_DEPTH); Removed negation in cfs_tx_ctrl.v: assign pop_ready = (pop_valid & md_tx_ready);
BUG-06	UVM_ERROR scoreboard(149) @16335ns — IRQ not received within 10 clock cycles	`cfs_regs.sv  (irq assign logic)` **Fix:** Added 5th missing OR condition: assign irq = (edge_max_drop & irqen_max_drop) | (4 other conditions).
BUG-07	Fatal error in cfs_algn_scoreboard:: write_in_agent_tx Line: 285	`cfs_md_monitor.sv  Line: 74` **Fix:** Changed '<' to '>=' in stuck threshold check: if (item.length >= agent_config.get_stuck_threshold())
BUG-08	RegModel mismatch: STATUS DUT=0x2 vs mirrored=0x1 (uvm_reg.svh:2889)	`cfs_algn_reg_block.sv cfs_regs.sv  (STATUS logic)` **Fix:** Changed STATUS offset from 'h0008 to 'h000C in default_map.add_reg. Removed '+1' from status_cnt_drop field assignment.

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BUG-09	UVM_ERROR predictor(120) @675ns Bus status mismatch: UVM_IS_OK vs UVM_NOT_OK (APB READ addr:8)	`cfs_algn_reg_predictor.sv Line: 120 cfs_algn_reg_block.sv` **Fix:** Root cause same as BUG-08: STATUS offset wrong. Resolved by correcting offset to 'h000C.
BUG-10	Incorrect offset in push_data cfs_rx_ctrl.v	`cfs_rx_ctrl.v (push_data offset field)` **Fix:** Changed md_rx_offset+1 to md_rx_offset in push_data offset field assignment.